Course Name: BCSC 0004: Operating System

447626

Course Outcome

CO1- Understand the classification of operating system environment.
CO2- Apply the concept of CPU process scheduling for the given scenarios
CO3- Illustrate the process synchronization and concurrency process in operating system

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Mid Term Examination, Odd Semester 2024-25
B.Tech (CS/AIML & IoT/BCT/CCV/CSF/DA), IInd, III Sem
BCSC 0004: Operating System

Time: 2 Hours Maximum Marks: 30

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Explain time sharing system with its benefits over multiprogramming.	3	1	R	F
2	Could you explain how semaphores can be used to enforce the sequencing of operations between threads, providing an example to illustrate your explanation?	3	2	An	P
3	A process is executing in the CPU when an interrupt occurs. The process needs to be suspended, and another process needs to be loaded into the CPU. How does the Operating System facilitate context switching between processes?	2+1	1	A	P
4	An operating system uses *shortest remaining time first* scheduling algorithm for pre-emptive scheduling of processes. Consider the following set of processes with their arrival times and CPU burst times (in milliseconds):	3	2	A	P
5	Explain race condition with the help of an example. Also, discuss a method to prevent race condition.	3	2	U	F

Process	Arrival Time	Burst Time
P1	0	12
P2	2	4
P3	3	6
P4	8	5

Find out the average average waiting time (in milliseconds) of the processes.

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Consider the following four processes with arrival times (in milliseconds) and their length of CPU bursts (in milliseconds) as shown below: Process P1 P2 P3 P4 Arrival time 0 3 5 6 CPU burst time 7 3 5 2 These processes are executed in such a manner that after each 2 millisecond there will be a context switching occur. All the process if follow Round Robin then calculate the average waiting time and average turn around time for the last arriving process.	5	2	A	P
7	A file processing application is designed to read data from a file, perform some calculations, and then write the processed results to a database. The process goes through several stages, interacting with the CPU, I/O devices, and memory. Due to system resource constraints and the need for I/O operations, the process may be temporarily suspended by the operating system. Using the appropriate diagram, explain the state transitions the file processing application undergoes during its lifecycle.	5	3	An	C
8	There **n** number of persons are waiting in queue in railway ticket booking/enquiry counter. The 1st person is asking for availability of trains, 2nd will be asking for booking seat, 3rd will again be asking for seat availability and so on. Now when you reach to the counter you see that there are big hassle in the queue and everyone want to ask for their turn first. From the concept of synchronization what is your approach to make this system hassle free.	5	3	An	M